

Checkpoint A | Version 3 | Solutions

Each question earns *Pass* or *Not Yet*. *Pass (high)* and *Pass (low)* mark the range of passing work; both earn the same *Pass*.

Q1 | Opportunity Cost (A1.1)

Suppose Arthur can make 6 cakes *or* 18 drinks in one day.

What to look for:

Pass (high): Both per-unit tradeoffs and the best alternative forgone for each choice.

Pass (low): Correct ratios with an arithmetic slip, or an omitted unit made clear by the work.

Not Yet: Reversed tradeoffs, adding all alternatives, or missing evidence of the best alternative.

Grading note: On this checkpoint, a response with either parts (a)–(b) or part (c) correct earned Pass. The wording was less clear than intended, so we resolved that in the student's favor. Future assessments of this skill expect both.

a) What is Arthur's opportunity cost of 1 cake? **3 drinks**

■ a) **3 drinks:** giving up 18 drinks buys 6 cakes, so one cake costs 18/6 drinks.

b) What is Arthur's opportunity cost of 1 drink? **1/3 cake**

■ b) **1/3 cake,** the reciprocal.

c) Arthur's options for the day, best first: take a Ministry shift, work in the shop, de-gnome the garden.

(i) What is his opportunity cost of working in the shop? **Taking a Ministry shift**

■ c) (i) The Ministry shift, his best alternative to the shop.

(ii) What is his opportunity cost of the Ministry shift? **Working in the shop**

■ c) (ii) Working in the shop, his best alternative to the shift. De-gnoming, ranked last, is the cost of nothing.

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Q2 | Arthur's PPF (A1.2)

Suppose Arthur can make 6 cakes *or* 18 drinks in one day. There are two graphs, one beside (a) and one beside (b). On both graphs, draw Arthur's PPF, drinks on the vertical axis and cakes on the horizontal. Label the intercepts.

What to look for:

Pass (high): A meaningful PPF, inside/on/outside classifications, and the effect of each capacity change.

Pass (low): An isolated wrong or missing label when the graph and other answers establish the distinction.

Not Yet: No usable PPF graph, repeated category confusion, or a wrong capacity-change answer without supporting evidence.

Grading note: On this checkpoint, a response with either part (a) or part (b) correct earned Pass. The wording was less clear than intended, so we resolved that in the student's favor. Future assessments of this skill expect both.

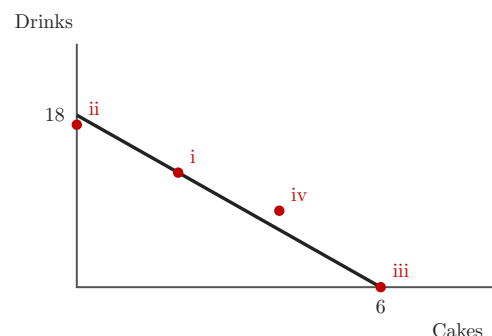
a) For each daily output, write whether it is unattainable, inefficient, or efficient, and mark it on the first graph.

(i) 2 cakes and 12 drinks **Efficient**

(ii) 17 drinks **Inefficient**

(iii) 6 cakes **Efficient**

(iv) 4 cakes and 8 drinks **Unattainable**

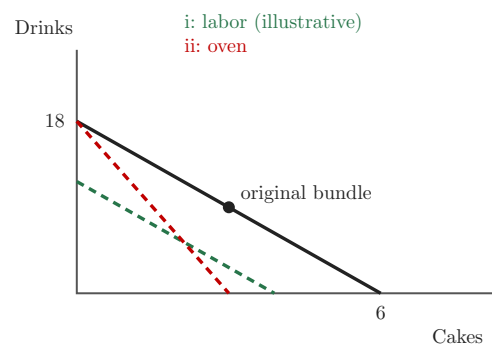


PPF: Straight line from 18 drinks to 6 cakes. Opportunity cost: 3 drinks per cake. a) **In order:** Efficient; Inefficient; Efficient; Unattainable.

b) Arthur currently makes 3 cakes and 9 drinks. On the second graph, draw his PPF and mark this output. Is that output unattainable, inefficient, or efficient:

(i) if he cuts back the labor he devotes to these two foods? **Unattainable**

(ii) if instead his oven breaks down, halving his daily output of cakes? **Unattainable**



b) The original bundle is on the original frontier. (i) Less labor: both intercepts shrink; **unattainable**. (ii) Oven: cake intercept falls to 3, drinks stay at 18; **unattainable**. **Taylor's Note:** In Q2.b, new frontiers illustrate the shifts but they are not required in the answer.

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Q3 | Absolute and Comparative Advantage (A2.1)

Luna and Neville both operate candy shops specializing in Tarts and Chocolates. In one day, Neville can make 12 tarts or 9 chocolates, and Luna can make 4 tarts or 8 chocolates.

What to look for:

Pass (high): Production and cost tables; AA from greater output and CA from lower opportunity cost.

Pass (low): A copying or arithmetic slip and its consistent consequences when the method and comparisons are clear.

Not Yet: Using output to choose CA, systematically reversing costs, or wrong answers with no cost work to show where the error came from. Correct answers earned Pass even without the cost work shown.

Set up a Production Table with both shops' output per day.

	Tart	Chocolate
Neville	12 Tarts	9 Chocolates
Luna	4 Tarts	8 Chocolates

Production table: each shop's output per day if it makes only that good, straight from the setup.

Then set up an Opportunity Cost table with both shops' opportunity costs for each good.

	Tart	Chocolate
Neville	$\frac{3}{4}$ Chocolate	$\frac{4}{3}$ Tarts
Luna	2 Chocolates	$\frac{1}{2}$ Tart

Opportunity costs: each cell is what the shop gives up of the other good. Neville: a tart costs $\frac{9}{12} = \frac{3}{4}$ chocolate, a chocolate costs $\frac{12}{9} = \frac{4}{3}$ tarts. Luna: a tart costs $\frac{8}{4} = 2$ chocolates, a chocolate costs $\frac{4}{8} = \frac{1}{2}$ tart.

Which shop has the absolute advantage (AA) and comparative advantage (CA) in each good?

AA in Tarts: **Neville**

CA in Tarts: **Neville**

AA in Chocolates: **Neville**

CA in Chocolates: **Luna**

AA in tarts: **Neville** ($12 > 4$). AA in chocolates: **Neville** ($9 > 8$).

CA in tarts: **Neville** ($\frac{3}{4} < 2$). CA in chocolates: **Luna** ($\frac{1}{2} < \frac{4}{3}$). Neville is better at both, but each shop holds exactly one comparative advantage.

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Q4 | Trade Between the Twins (A3.1)

Fred and George each make Fireworks and Pastilles for their joke shop. Their opportunity costs are listed in the following opportunity cost table.

	Firework	Pastille
Fred	4 Pastilles	1/4 Firework
George	3/2 Pastilles	2/3 Firework

What to look for:

Endpoints: The bounds identify an open interval. At an endpoint one producer is indifferent.

Pass (high): The lower-cost specialist, the full trade interval in common units, and an acceptable rate.

Pass (low): Correct specialist, bounds, and acceptable selection plus one extra outside checkbox.

Not Yet: Wrong specialization or bounds without correct work, only outside rates selected, or multiple extra selections.

a) If the twins specialize and trade with each other, which one should specialize in Pastilles? **Fred**

a) **Fred**, the twin with the lower opportunity cost of a pastille. George has the lower cost of a firework (3/2 versus 4 pastilles) and Fred the lower cost of a pastille (1/4 versus 2/3 firework).

b) What is the range of exchange rates that would make both Fred and George better off?

Between **3/2** and **4** Pastilles for 1 Firework

b) Between **3/2** and **4** pastilles for 1 firework: the two twins' opportunity costs of a firework, read down the Firework column. Below the low end the firework maker would rather keep his fireworks; above the high end the pastille maker would rather make fireworks himself.

c) Select all the exchange rates which could facilitate a trade between the twins.

☐ 1 Firework for 1 Pastille ☒ 1 Firework for 2 Pastilles ☐ 1 Firework for 5 Pastilles ☐ 1 Firework for 6 Pastilles

c) Only **1 firework for 2 pastilles**. 1 is below George's cost of 3/2, so he keeps his fireworks; 5 and 6 are above Fred's cost of 4, so he would rather make fireworks himself.